G.3 Comparison of Candidate Trajectory-Prediction Algorithms

The highlighted row is implemented. The comparison is qualitative and does not claim measured accuracy equivalence.

Algorithm	Core idea	History / training need	Online cost	Noise and turn behaviour	Limitation	Role in this project
Last observation carried forward	Hold the last position.	One observation; no training.	Negligible.	No noise growth, but no motion or turn representation.	Moving aircraft appears stationary.	Emergency display fallback only.
Constant velocity and heading	Advance position using speed, heading and vertical rate.	One recent state; no training.	Very low.	Sensitive to noisy motion fields and cannot anticipate turns.	Error grows during manoeuvres.	Deterministic fallback when regression is unavailable.
Sliding-window linear regression (USED)	Fit latitude, longitude and altitude against time.	At least 4 and up to 20 samples; no offline training.	Low.	Averages several observations but assumes locally straight motion.	Weak during sustained turns or acceleration.	Transparent baseline for short receiver gaps.
Kalman filter [10]	Update a dynamic state using prediction and measurement steps.	Sequential observations; no labelled training set.	Low to moderate.	Explicitly models measurement noise; turn handling depends on state model.	Needs process and covariance tuning.	Most practical next real-time upgrade.
Polynomial regression	Fit a curved function of position versus time.	Several recent samples; no offline training.	Low to moderate.	Can follow gentle curvature but may amplify noise outside the sample window.	Extrapolation can become unstable.	Research comparison with a tightly limited horizon.
Random forest / boosting [11]	Learn nonlinear corrections from engineered trajectory features.	Representative labelled trajectories required.	Moderate.	Can learn nonlinear patterns represented in training data.	Does not naturally model time unless features encode it.	Offline comparative study after a verified dataset is collected.
LSTM / GRU [12]-[14]	Learn temporal dependencies from trajectory sequences.	Large, diverse sequential dataset required.	Moderate to high.	Can represent complex manoeuvres if training coverage is sufficient.	Data-, compute- and validation-intensive.	Future large-dataset work, not justified for the current prototype.

Table 15: Trajectory Prediction Algorithm Comparison